

Applying a Stability-Quantification Method in Complex Networks to Epileptic Mouse Brains

Gilmar Jose Alves de Souza Junior

SID: 530191337

Supervisor: Prof. Joseph T. Lizier

School of Computer Science & Centre for Complex Systems, The University of Sydney

CSYS5050 Complex Systems Capstone — June 2026

Abstract

Focal epilepsy is a network disorder: seizures ignite in a localised epileptogenic zone and recruit the brain through its structural wiring. Yet the standard structural diagnostic of dynamical stability—the dominant eigenvalue of the coupling matrix—is binary, discards the sub-dominant spectrum, and offers no node-level resolution. This project applies the recently introduced *deviation-from-stability* (D_{st}) framework of Liao and Lizier, and its two node-level centralities (stability *susceptibility* $D(\rightarrow i)$ and *influence* $D(k \rightarrow)$), to DTI-derived structural connectomes of six mice: two from an epileptic case cohort (ANDERSON) and four controls (ARNOLD), each parcellated into 66 regions. Working entirely from the connectome—with no seizure recordings—we find that (i) network-level D_{st} under a symmetry-breaking column normalisation shows non-overlapping cohort ranges (min case 0.792 > max control 0.788; $p = 0.080$); (ii) a purely structural centrality reproduces an independent dynamics-based ground truth, the per-node critical excitability $x_{0,i}^c$ derived from an Epileptor model, with a near one-to-one relationship in both case mice; (iii) the choice of normalisation, not parameter tuning, decides whether the centralities carry independent information; and (iv) per-region and left–right contrasts recover epileptologically interpretable structures (gustatory cortex, hippocampal formation, sensorimotor and olfactory cortex). The work extends a prior single-patient result to a cohort across two strains and adds a hemispheric-asymmetry analysis. The strain effects are promising but underpowered at $n = 2$ vs 4; we report effect sizes alongside significance and treat the column scheme as primary rather than universal.

Contents

1	Introduction	2
2	Background	3
2.1	Epilepsy as a network disorder and the Epileptor	3
2.2	Limits of the dominant-eigenvalue criterion	3
2.3	Deviation from stability and its node decomposition	3
3	Methodology	4
3.1	Data	4
3.2	From connectome to coupling matrix C	4
3.3	Stability centralities and D_{st}	5
3.4	Critical excitability $x_{0,i}^c$	5
3.5	Cohort statistics	5

4 Results and Discussion	5
4.1 All mice are healthy, and network D_{st} separates the cohorts under <code>column</code>	6
4.2 A structural centrality reproduces a dynamics-based ground truth	7
4.3 Whether D adds information depends on the normalisation	7
4.4 Per-region outliers flag epileptologically interpretable structures	8
4.5 A left–right asymmetry signal beyond the single-patient study	9
5 Limitations and future work	10
6 Conclusion	10
A Acknowledgement of generative-AI use	12
B Reproducibility	12
C Full result figures	12
C.1 Network-level D_{st} across the cohort	12
C.2 Per-region stability centralities	13
C.3 Centrality correlations	15
C.4 Anderson vs Arnold per-region contrasts	18
C.5 Left–right asymmetry	20
C.6 Cohort comparison figures (<code>column</code>)	21
D Full result tables	21
D.1 Network D_{st} and spectral radius $\rho(C)$	22
D.2 Top regions by stability susceptibility	22
D.3 Anderson vs Arnold per-region outliers	24
D.4 Systematic laterality	25
D.5 Homotopic-pair laterality tests	25

1 Introduction

Epilepsy affects roughly 1 in 100 people (Fiest et al., 2017). In *focal* epilepsy, seizures begin in a circumscribed *epileptogenic zone* (EZ) and then spread to the rest of the brain along its anatomical connections. Locating that zone is the central clinical question, but it is usually inferred only after weeks of invasive intracranial monitoring (Rosenow & Lüders, 2001). This motivates a complementary, non-invasive question that is squarely a complex-systems problem: *can a measure computed entirely from the structural connectome—with no seizure recordings—rank brain regions by how close they sit to a seizure, and tell an epileptic cohort apart from controls?*

A network of diffusively coupled dynamical units is linearly stable around its resting state when every eigenvalue of the effective coupling matrix C has real part below one. The textbook diagnostic is the dominant eigenvalue λ_{max} —a single scalar. As an early-warning measure it has three weaknesses. It is *binary* (stable vs. unstable) and says nothing about how close to the edge a network sits; it *ignores the rest of the spectrum*, even though a real perturbation excites all modes; and it has *no node-level decomposition*, so it cannot say which region drives instability and therefore cannot localise an EZ (Liao & Lizier, 2026).

Liao & Lizier (2026) close this gap with a measure of *deviation from stability*, D_{st} , that is continuous, accounts for the full spectrum, and decomposes onto individual nodes as two *dynamical* centralities. Their companion thesis (Liao, 2026) applied the framework to a single human epilepsy patient and showed that the node-level centralities track an independent, dynamics-based notion

of seizure proximity. The present capstone tests whether that link is a property of brain networks in general rather than of one subject, by (1) adopting the principled D_{st} framework, (2) scoring every region of real mouse connectomes with its two centralities, and (3) testing the signal on a real cohort of two epileptic and four control animals. We add two analyses absent from the single-patient thesis—a cross-strain contrast and a left–right asymmetry test—and we make explicit how the conclusions depend on one load-bearing modelling choice, the connectome normalisation.

2 Background

2.1 Epilepsy as a network disorder and the Epileptor

Computational epileptology increasingly treats seizures as a whole-brain network phenomenon. The *Epileptor* (Jirsa et al., 2014) is a canonical neural-mass model whose slow variable governs the transition into seizure as an excitability parameter x_0 is raised; embedded on a structural connectome it underlies the Virtual Epileptic Patient pipeline (Jirsa et al., 2017) and The Virtual Brain (Sanz Leon et al., 2013). These models simulate seizure spread; our question is the dual one of *stability*: how far the resting network sits from losing stability, read directly from structure.

2.2 Limits of the dominant-eigenvalue criterion

Linear stability analysis around a fixed point reduces to the spectrum of C . Using only λ_{max} throws away information that an early-warning measure needs: the graded distance to the stability boundary and the contribution of every sub-dominant mode. Three established families of method bear on this problem, each with a complementary limitation. Whole-brain simulators such as the Virtual Epileptic Patient (Jirsa et al., 2017; Sanz Leon et al., 2013) reproduce seizure *spread* mechanistically, but need patient-specific tuning and forward simulation and speak to propagation rather than proximity to instability. Effective-connectivity methods—transfer entropy (Vicente et al., 2011) or dynamic causal modelling (Friston et al., 2003)—recover genuinely *directed* interactions, but require functional recordings a structural connectome cannot supply. Classical graph centralities (Rubinov & Sporns, 2010; Liao & Lizier, 2026) are read straight from that connectome yet score topological prominence, not dynamical response. D_{st} targets the gap these leave—continuous, spectrum-wide, and node-decomposable from structure alone—at the explicit cost of *imposing* rather than measuring edge direction (§5).

2.3 Deviation from stability and its node decomposition

Following Liao & Lizier (2026), drive every node of the linearised system with small independent noise. The fluctuations $X(t) = z(t) - z^*$ obey a multivariate Ornstein–Uhlenbeck process

$$dX(t) = -X(t) (I - C) \theta dt + \zeta dW(t), \quad (1)$$

with reversion rate θ and noise scale ζ (both set to 1 here) and W a Wiener process with identity covariance. The system settles to a stationary covariance Ω , and the deviation from stability is its per-node average,

$$D_{\text{st}} = \frac{1}{N} \text{trace}(\Omega). \quad (2)$$

Ω is obtained *analytically*—no simulation—as a power series in C , valid for $\rho(C) < 1$ (Liao & Lizier, 2026; Schwarze & Porter, 2021):

$$\Omega = \frac{\zeta^2}{2\theta} \sum_{m=0}^{\infty} 2^{-m} \sum_{u=0}^m \binom{m}{u} (C^u)^\top C^{m-u}. \quad (3)$$

Crucially, D_{st} equals a weighted sum of *convergent dual-walk motifs*: pairs of directed walks that start at a common source k and reinforce on a common target i . More convergent walks means perturbations pile up and the network sits further from stability. Regrouping Eq. (3) by target node i or source node k gives the two node-level centralities,

$$D(\rightarrow i) = \Omega_{ii} \quad (\text{stability susceptibility}), \quad D(k \rightarrow) = (\Omega_{C^\top})_{kk} \quad (\text{stability influence}). \quad (4)$$

$D(\rightarrow i)$ measures how easily node i is driven from rest—“downstream of a lot”—and $D(k \rightarrow)$ measures node k ’s capacity to destabilise others—“upstream of a lot”. Unlike eigenvector centrality these measure stability *directly*, and for a symmetric C they coincide. This distinction is the lever exploited below.

3 Methodology

3.1 Data

Each mouse contributes one DTI-derived structural connectome, coarse-grained to $N = 66$ brain-atlas regions and supplied as a symmetric 66×66 weighted adjacency matrix K_{raw} (symmetry error $\|K - K^\top\|_{\text{max}} \sim 10^{-12}$, float-roundoff level). The symmetry is intrinsic to diffusion tractography, which cannot resolve edge direction and is prone to false-positive connections (Maier-Hein et al., 2017). The operational cohort is six animals—an epileptic case cohort ANDERSON_1, ANDERSON_2 and four controls ARNOLD_2 – ARNOLD_5—provided by Prof. Daria Anderson. Atlas placeholder rows (*BACKGROUND, *_MASK) carry no physical connectivity and are flagged trivial; they are excluded from rankings, from the critical-excitability sweep, and from the multiple-testing family. Region labels use nominal L-/R- prefixes; because the underlying hemispheric intervention may not align with anatomical left, we treat the sides as nominal throughout.

3.2 From connectome to coupling matrix C

Turning K_{raw} into the coupling matrix C is the most consequential modelling step, so we run three normalisation schemes side by side (Table 1).

- **tvb** (Epileptor reference): zero the diagonal, truncate edges above the 95th percentile, and rescale to $[0, 1]$, matching the TVB pipeline (Sanz Leon et al., 2013). We then embed a one-dimensional Epileptor on each node, solve the all-healthy fixed point ($x_{0,i} = -2.3$) with `fsolve`, and take C as the discretised Jacobian. Symmetric, and the only scheme admitting the critical-excitability sweep (§3.4).
- **parkes** (literature bridge): $K = A/(\rho(A) + c)$ with $c = 1$, the prescription of Parkes et al. (2024), which pins $\rho(C)$ just below one. Symmetric; $C = K$ directly.
- **column** (primary): rescale every column sum to 0.95. This *breaks* the input symmetry, so $D(\rightarrow i) \neq D(k \rightarrow)$ in general; $C = K$ directly.

For the two linear schemes the Epileptor solve is bypassed and the normalised K is used as C , because those normalisations encode synaptic gain directly. Importantly, lowering the Parkes constant c to push $\rho(C) \rightarrow 1$ does *not* decouple $D(\rightarrow i)$ from eigenvector centrality: for symmetric C , $\Omega = \frac{\zeta^2}{2}(I - C)^{-1}$ and c rescales the spectrum uniformly, leaving eigenvectors and spectral-gap ratios invariant. Breaking symmetry—what **column** does—is the right knob. The **column** scheme is not claimed as a biological model of directionality; it is used as a controlled symmetry-breaking probe to test whether the D_{st} decomposition can carry node-specific information once the degeneracy of symmetric structural input is removed.

Table 1: Normalisation schemes exercised side by side. Symmetric input $\Rightarrow D(\rightarrow i) \equiv D(k \rightarrow)$ at machine precision for symmetry-preserving schemes.

Scheme	Role	Symmetric C ?	$x_{0,i}^c$ sweep?	$\rho(C)$ (cohort)
tvb	Epileptor reference	Yes	Yes	0.99915
parkes	Literature bridge	Yes	No	≈ 0.99999
column	Primary	No	No	0.95 (exact)

3.3 Stability centralities and D_{st}

For every mouse and scheme we first confirm the healthy regime $\rho(C) < 1$ —the precondition for Eq. (3) to converge—then compute Ω via the power series (iteration cap 10^8), and read off $D(\rightarrow i)$, $D(k \rightarrow)$ (Eq. (4)) and D_{st} (Eq. (2)). The continuous-time OU model (Eq. (1), `discreteTime=false`) is used throughout. We also compute ten classical centralities (in/out-degree, left/right eigenvector, PageRank, Katz, self-communicability, betweenness, in/out-closeness) on the same C for comparison (Rubinov & Sporns, 2010).

3.4 Critical excitability $x_{0,i}^c$

For the Epileptor scheme only, we obtain an *independent*, dynamics-based notion of seizure proximity. For each non-trivial node i we raise its excitability $x_{0,i}$ alone—all others held at the healthy baseline -2.3 —until the network loses linear stability ($\rho(C) \geq 1$), via a coarse forward sweep (step 0.01) followed by bisection (tolerance 10^{-4}). The resulting *critical excitability* $x_{0,i}^c$ is the smallest single-node excitability that destabilises the whole network; a *lower* $x_{0,i}^c$ means a region is closer to its own seizure threshold. Because $x_{0,i}^c$ comes from the nonlinear Epileptor bifurcation while $D(\rightarrow i)$ comes from the linear covariance, agreement between them is a genuine cross-method check rather than a tautology.

3.5 Cohort statistics

Network level. `compareDstAcrossCohort` classifies each mouse by name prefix and runs a Welch t -test (unequal variance) on D_{st} between strains. **Per region.** For each case mouse a and metric $M \in \{D(\rightarrow i), D(k \rightarrow), \text{BC}\}$ we form a z -score against the control mean and SD, $z_{a,i} = (x_{a,i} - \mu_{\text{Arn},i}) / \sigma_{\text{Arn},i}$, convert to two-sided p -values, and flag outliers by *Bonferroni* correction over the family of (non-trivial nodes $\times$ metrics) at $\alpha = 0.05$. Bonferroni is conservative but transparent and, with only $n_{\text{Arn}} = 4$, more defensible than FDR given the noisy per-node SD. **Left-right.** For each homotopic pair p we compute the laterality index $\text{LI}_{\text{norm}} = (L - R) / (L + R) \in [-1, 1]$ and the signed difference $\text{LI}_{\text{signed}} = L - R$. Three views follow: per-case z -scores vs. the control band (Bonferroni-flagged); a per-pair Welch t -test between strains; and a cohort-wide sign-rank test of LI_{norm} against zero to detect a systematic side bias. The full pipeline, parameters, and provenance for all three scheme runs are recorded automatically and are reproducible from a single configuration file.

4 Results and Discussion

Table 2 consolidates the cross-scheme picture the rest of this section develops: the network-level cohort signal and its significance, the mutual redundancy of the two stability centralities and their redundancy with eigenvector centrality, and whether an independent dynamical validation exists. One qualitative axis organises every row—symmetry of C : the centralities are degenerate under **parkes**/**tvb** and decouple only under **column**.

Table 2: Cross-scheme results summary (six mice: two ANDERSON cases, four ARNOLD controls). D_{st} entries are cohort means, case/control; absolute D_{st} scales with the normalisation, so only the *within*-scheme contrast (pattern and Welch p) is comparable across rows, and no scheme reaches $p < 0.05$ at this sample size. Only **column** yields disjoint cohort ranges with the case cohort *less* stable; under **parkes** the case mean even sits *below* the controls. $\rho(D(\rightarrow i), D(k \rightarrow))$ and $\rho(D(\rightarrow i), \text{EC})$ are cohort-mean Spearman correlations, and “ $\equiv$ ” marks values identical by construction for symmetric C . Rows mirror Table 1; see Figs. 1, 2, 3.

Scheme	Cohort D_{st} (case/ctrl)	Welch p	$\rho(D(\rightarrow i), D(k \rightarrow))$	$\rho(D(\rightarrow i), \text{EC})$	$x_{0,i}^c$ validation
tvb	Overlap, 273.4/270.6	0.267	1.00 ($\equiv$)	$\approx +1.0^b$	Yes
parkes	Overlap, 599.2/623.9	0.739	1.00 ($\equiv$)	$\approx +1.0$	No
column	Separated , 0.800/0.773	0.080	0.75	+0.76 / -0.77^a	No

^a Out- vs in-eigenvector centrality under the symmetry-broken scheme; the sign-split *is* the in/out decoupling that lets $D(\rightarrow i)$ and $D(k \rightarrow)$ carry distinct per-node information.

^b Collapse onto EC holds on the non-trivial nodes, as for **parkes**; the raw cohort figure (App. C) reads -1.0 only because the placeholder rows—excluded from every ranking (Methods)—inflate $D(\rightarrow i)$ under **tvb**.

4.1 All mice are healthy, and network D_{st} separates the cohorts under **column**

Every mouse sits in the healthy regime under all three schemes ($\rho(C) \leq 0.99999$), so the centrality analysis is well-posed throughout. At the network level, D_{st} separates the strains only under the symmetry-breaking **column** scheme (Fig. 1): the case ranges and control ranges do not overlap, with the smallest case value (0.792) above the largest control value (0.788), and the epileptic cohort globally *less* stable (higher D_{st}). The Welch test is $t = 2.85$, $p = 0.080$ —non-significant at $n = 2$ vs 4, so the clean range separation is best read as a hypothesis, not a settled result (§5). Under **parkes** and **tvb** the cohort D_{st} ranges overlap entirely. That the separation appears *only* when symmetry is broken is the first sign that the symmetry-breaking knob, not the choice of constant, is doing the work. Biologically the direction is the expected one—the epileptic cohort is the *less* stable, sitting closer to losing stability—so, should it survive a larger sample, the range separation is a candidate cohort-level signature of the epileptic strain rather than a curiosity of six animals.

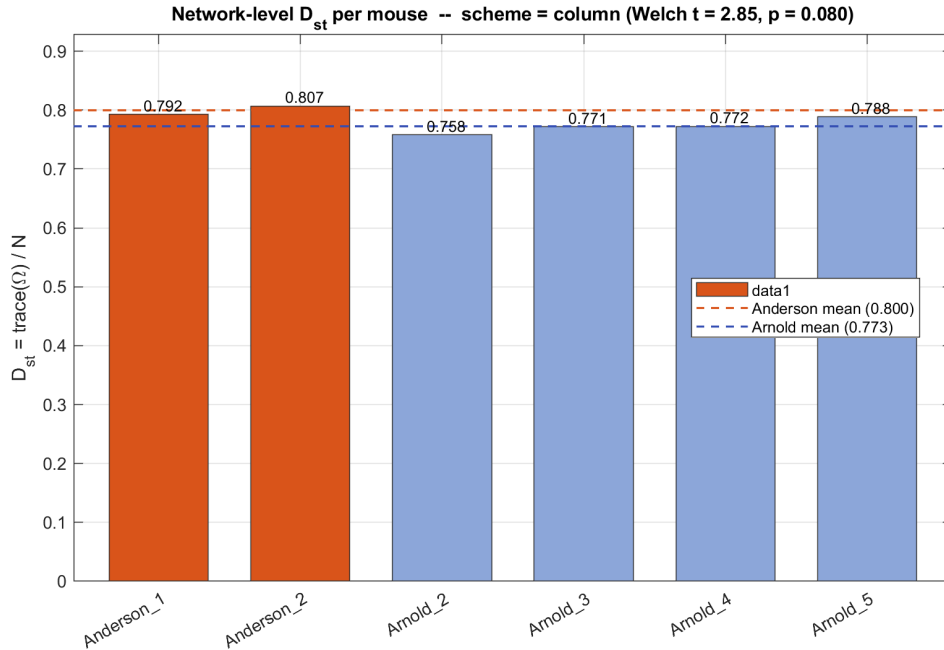


Figure 1: Network-level $D_{\text{st}} = \text{trace}(\Omega)/N$ per mouse, **column** scheme. Case (ANDERSON, orange) and control (ARNOLD, blue) ranges do not overlap (min case 0.792 > max control 0.788); dashed lines are cohort means (0.800 vs 0.773). Welch $t = 2.85$, $p = 0.080$.

4.2 A structural centrality reproduces a dynamics-based ground truth

Under the Epileptor (**tvb**) scheme, per-region stability centrality and the independently computed critical excitability $x_{0,i}^c$ trace a near one-to-one decreasing curve in both case mice (Fig. 2): regions with the highest $D(\rightarrow i)$ are exactly those that destabilise the network at the lowest added excitability—i.e. those closest to their own seizure threshold. The same regions sit atop the curve in **ANDERSON_1** and **ANDERSON_2**, and the consensus top regions are high-connectivity thalamic and basal-ganglia nuclei. This reproduces the single-patient result of Liao (2026) across animals: a purely *structural* score recovers a *dynamical* ground truth, and it does so consistently, which validates the method itself before any epilepsy-specific contrast. Because $x_{0,i}^c$ is read from the nonlinear Epileptor bifurcation while $D(\rightarrow i)$ is read from the linear covariance, the two derivations are methodologically independent; their agreement across both case mice is strong evidence that D is consistent with a biologically meaningful network property rather than an artefact of either route.

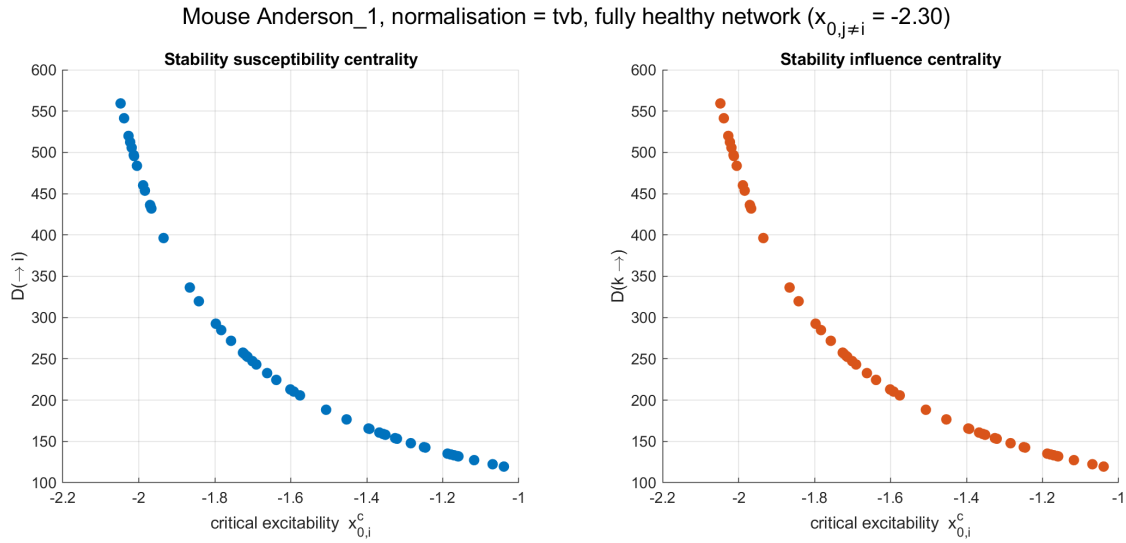


Figure 2: Per-region stability centrality vs. critical excitability $x_{0,i}^c$ for **ANDERSON_1** (**tvb**). Left: susceptibility $D(\rightarrow i)$; right: influence $D(k \rightarrow)$ (identical under the symmetric scheme). High centrality corresponds to low $x_{0,i}^c$, i.e. proximity to the seizure threshold. The relationship replicates in **ANDERSON_2**.

4.3 Whether D adds information depends on the normalisation

Under the symmetry-preserving normalisations (**parkes**, **tvb**), the node-level centralities prove largely redundant, collapsing onto a single ranking. Across the cohort the walk-based measures—weighted in/out-degree, eigenvector centrality, PageRank, Katz, self-communicability, and the stability susceptibility $D(\rightarrow i)$ —agree at cohort-mean Spearman $\rho \geq 0.98$ (median 0.99); even shortest-path closeness tracks them at $\rho \approx 0.96$, leaving betweenness centrality (Rubinov & Sporns, 2010) as the only meaningful departure ($\rho \approx 0.79$ against $D(\rightarrow i)$). Two mechanisms drive the collapse. *First*, symmetry of the coupling matrix forces the two directed stability centralities to coincide exactly: the susceptibility $D(\rightarrow i)$ and the influence $D(k \rightarrow)$ are then read from one and the same stationary covariance $\Omega = \frac{\zeta^2}{2}(I - C)^{-1}$ (Liao & Lizier, 2026). *Second*, the connectomes carry a clear spectral gap—the ratio $\lambda_2/\lambda_{\max}$ lies between 0.49 and 0.68 across the six mice—so the leading eigenvalue is well separated from the rest. In the spectral form $\Omega = \frac{\zeta^2}{2} \sum_k (1 - \lambda_k)^{-1} q_k q_k^\top$ the leading mode q_1 then dominates the node-to-node variation of Ω , and because every walk-based measure is a weighted sum over powers of C (Schwarze & Porter, 2021) all inherit that dominant-mode profile and reduce to the leading eigenvector—that is, to eigenvector centrality (indeed $D(\rightarrow i)$ correlates with it at $\rho \approx 0.997$). The system moreover sits just *below* the stability boundary ($\rho(C) \approx 0.99999$ under **parkes**), which weights q_1 by

$(1 - \lambda_{\max})^{-1}$ —more than four orders of magnitude above the next mode—so that the leading mode accounts for $\approx 99.9\%$ of $\text{trace}(\Omega)$ and Ω is, for ranking purposes, effectively of rank one. Betweenness escapes not merely because it is path-based—closeness is too, yet still collapses—but because it scores how often a node bridges shortest paths between *other* regions, a bottleneck role (here flagging white-matter, CSF and cranial-nerve nodes) orthogonal to dominant-mode magnitude.

The redundancy is therefore a deterministic consequence of symmetry and spectral structure, not an artefact of the small sample size, and it cannot be relieved by rescaling the coupling strength: scaling keeps C symmetric, so $D(\rightarrow i) \equiv D(k \rightarrow)$ exactly, and preserves both the eigenvectors and the gap ratio $\lambda_2/\lambda_{\max}$, leaving $\rho(D(\rightarrow i), \text{EC}) \geq 0.9996$ even as $\rho(C)$ is lowered to ≈ 0.99 (Parkes et al., 2024). It is relieved only by breaking the matrix symmetry (Fig. 3): under the **column** normalisation the two stability centralities decouple ($\rho \approx 0.75$), the in-style measures sign-flip against $D(\rightarrow i)$ ($\rho \approx -0.78$), and the susceptibility’s correlation with the out-eigenvector centrality falls to $\rho \approx 0.76$, recovering the independent directed signal on which the region-level comparison of susceptibility and influence depends. This is precisely the symmetric $\rightarrow$ asymmetric transition reported by Liao & Lizier (2026), here reproduced on real brain networks.

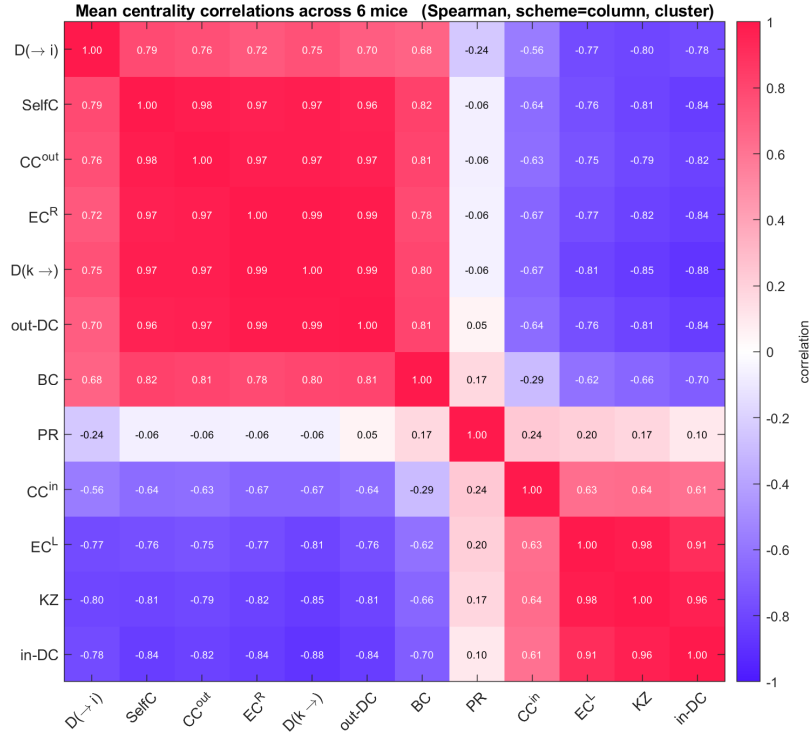


Figure 3: Cohort-mean Spearman correlation of $D(\rightarrow i)$ and $D(k \rightarrow)$ with ten classical centralities, **column** scheme. In-style measures anti-correlate with $D(\rightarrow i)$ and the two stability centralities decouple ($\rho \approx 0.75$), confirming D carries independent per-node information only when symmetry is broken.

4.4 Per-region outliers flag epileptologically interpretable structures

Contrasting each case mouse against the four-control band under **column** flags Bonferroni-significant susceptibility excesses concentrated in limbic and sensorimotor cortex (Fig. 4, Fig. 5). Two regions are *cohort-robust*, flagged in both case mice: **gustatory cortex (L)** (+20.6%, $z = 7.9$ in ANDERSON_2; +10.5%, $z = 4.0$ in ANDERSON_1) and the **hippocampal formation** (+9.9%, $z = 6.7$ on the L side of ANDERSON_2; +6.6%, $z = 5.4$ on the R side of ANDERSON_1). Single-mouse signals add somatosensory cortex (+12.3%), temporal-association cortex (+7.2%) and basal-ganglia striatum (+5.5%). Mesial temporal lobe epilepsy with hippocampal sclerosis is a canonical focal epilepsy syndrome (Wieser, 2004), so the hippocampal formation’s appearance as

a cohort-robust outlier is biologically reassuring. By contrast, classical betweenness on the same mice flags white-matter, CSF and cranial-nerve nodes—anatomical artefacts of the tractography rather than epileptogenic tissue—underlining that a dynamics-based centrality targets more plausible structures than a purely topological one.

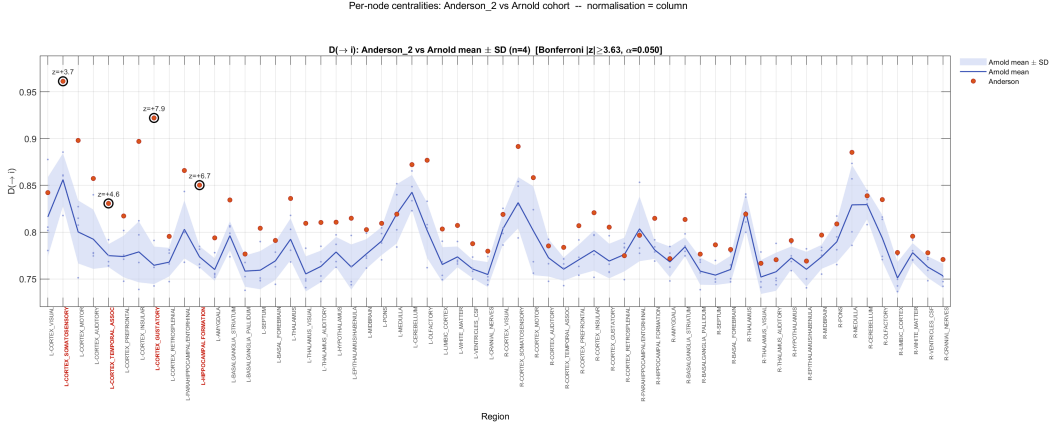


Figure 4: Per-region susceptibility $D(\rightarrow i)$ for ANDERSON_2 (red) against the four-control band (mean $\pm$ SD), column scheme. Ringed, bold-labelled nodes are Bonferroni-significant. Limbic and sensory cortical regions exceed the control band.

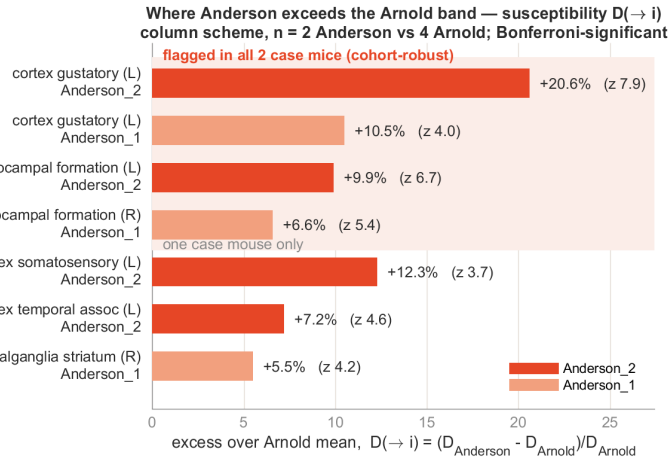


Figure 5: All Bonferroni-significant $D(\rightarrow i)$ excesses over the control mean (column). Gustatory cortex (L) and hippocampal formation flag in both case mice (shaded, “cohort-robust”); remaining rows are single-mouse signals.

4.5 A left–right asymmetry signal beyond the single-patient study

Laterality requires a cohort to define and is therefore absent from the thesis. At the cohort level, the case mice show a significant systematic side bias in LI_{norm} under every scheme (sign-rank $p < 0.002$), toward the L side under **column/parkes** and the R side under **tvb**; controls are largely silent. Per homotopic pair, the group-level Welch test flags the parahippocampal/entorhinal influence pair under **column** ($t = 4.50$, $p_{\text{bonf}} = 0.021$) and the white-matter pair under **tvb** ($p_{\text{bonf}} = 0.034$). The per-case view (Fig. 6) shows excess left-side *influence* at olfactory (+1.52, $z = 4.9$) and motor cortex (+0.75, $z = 6.3$) in ANDERSON_2; no homotopic pair survives

correction in ANDERSON_1, so this is suggestive rather than replicated. The limbic–thalamic flavour of these effects is consistent with the clinical relevance of left–right lateralisation in temporal-lobe epilepsy (Cendes et al., 1997; Li et al., 2000), with the caveat that the side labels are nominal. Surfacing a hemispheric signal at all is a cohort-only contribution beyond the single-patient study; at $n = 2$ cases, though, it is a lead to pursue with directed inputs rather than a settled asymmetry.

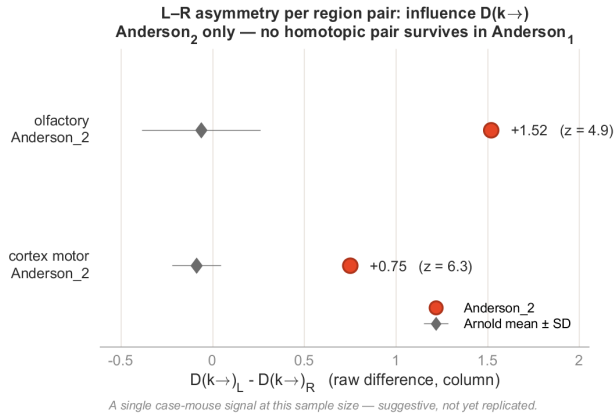


Figure 6: Left–right asymmetry of influence $D(k \rightarrow)$ per homotopic pair, **column**. Case value $L - R$ (red) vs. the control band (grey, mean $\pm$ SD). ANDERSON_2 shows excess left-side influence at olfactory and motor cortex; no pair survives correction in ANDERSON_1.

5 Limitations and future work

The cohort is small: $n = 2$ case vs 4 control. The D_{st} separation is therefore a hypothesis ($p = 0.080$) until replicated, and at this sample size only the largest effects clear Bonferroni correction—motor, amygdala and the laterality pairs sit near or below threshold for reasons of statistical power, not biology. We mitigate this by reporting effect sizes alongside z -scores and by leaning on the largest, replicated effects (gustatory cortex, hippocampal formation). The headline findings live in the **column** scheme; **tvb** and **parkes** tell a flatter story, so the conclusions are explicitly conditional on a modelling choice that we state rather than hide. Finally, the input is symmetric structural connectivity, so the in/out asymmetry that makes the centralities informative is imposed by normalisation rather than measured from directed interactions.

The natural next steps are: larger cohorts per strain to convert the D_{st} separation from hypothesis to result; *directed* effective connectivity (e.g. transfer entropy (Vicente et al., 2011) or DCM (Friston et al., 2003)) as input, so that $D(\rightarrow i)$ and $D(k \rightarrow)$ separate for a measured rather than imposed reason; and per-node critical-excitability thresholds computed under the primary **column** scheme, bridging the structural centralities and the dynamical ground truth within a single normalisation.

6 Conclusion

D_{st} gives a graded, full-spectrum stability signal where the dominant eigenvalue gives only a yes/no answer, and its two node-level centralities rank regions by a structural stability proxy plausibly related to epileptogenic vulnerability. On real mouse connectomes those rankings match an independent Epileptor-derived threshold and recover known seizure-onset biology; at

the network level, D_{st} shows non-overlapping cohort ranges in this small sample under column normalisation and reveals a new hemispheric-asymmetry signal absent from the single-patient study it builds on. Crucially, the framework’s usefulness on structural data hinges on breaking the connectome’s symmetry, not on tuning constants. With larger cohorts and directed inputs, deviation-from-stability centralities are a promising structural screening tool for prioritising candidate epileptogenic regions.

References

- Cendes, F., Caramanos, Z., Andermann, F., Dubeau, F., & Arnold, D. L. (1997). Proton magnetic resonance spectroscopic imaging and magnetic resonance imaging volumetry in the lateralization of temporal lobe epilepsy: a series of 100 patients. *Annals of Neurology*, 42(5), 737–746.
- Fiest, K. M., et al. (2017). Prevalence and incidence of epilepsy: a systematic review and meta-analysis of international studies. *Neurology*, 88(3), 296–303.
- Friston, K. J., Harrison, L., & Penny, W. (2003). Dynamic causal modelling. *NeuroImage*, 19(4), 1273–1302.
- Jirsa, V. K., Stacey, W. C., Quilichini, P. P., Ivanov, A. I., & Bernard, C. (2014). On the nature of seizure dynamics. *Brain*, 137(8), 2210–2230.
- Jirsa, V. K., et al. (2017). The Virtual Epileptic Patient: individualized whole-brain models of epilepsy spread. *NeuroImage*, 145, 377–388.
- Li, L. M., Caramanos, Z., Cendes, F., Andermann, F., Antel, S. B., Dubeau, F., & Arnold, D. L. (2000). Lateralization of temporal lobe epilepsy (TLE) and discrimination of TLE from extra-TLE using pattern analysis of magnetic resonance spectroscopic and volumetric data. *Epilepsia*, 41(7), 832–842.
- Liao, J. (2026). *Stability of Brain Networks in Epilepsy*, Ch. 4. Unpublished MSc thesis, The University of Sydney.
- Liao, J., & Lizier, J. T. (2026). Directly quantifying deviation from stability for linear dynamical systems on networks. Unpublished manuscript, School of Computer Science & Centre for Complex Systems, The University of Sydney.
- Maier-Hein, K. H., et al. (2017). The challenge of mapping the human connectome based on diffusion tractography. *Nature Communications*, 8, 1349.
- Parkes, L., Kim, J. Z., Stiso, J., et al. (2024). A network control theory pipeline for studying the dynamics of the structural connectome. *Nature Protocols*, 19, 3721–3749.
- Rosenow, F., & Lüders, H. (2001). Presurgical evaluation of epilepsy. *Brain*, 124(9), 1683–1700.
- Rubinov, M., & Sporns, O. (2010). Complex network measures of brain connectivity: uses and interpretations. *NeuroImage*, 52(3), 1059–1069.
- Sanz Leon, P., et al. (2013). The Virtual Brain: a simulator of primate brain network dynamics. *Frontiers in Neuroinformatics*, 7, 10.
- Schwarze, A. C., & Porter, M. A. (2021). Motifs for processes on networks. *SIAM Journal on Applied Dynamical Systems*, 20(4), 2516–2557.
- Vicente, R., Wibral, M., Lindner, M., & Pipa, G. (2011). Transfer entropy—a model-free measure of effective connectivity for the neurosciences. *Journal of Computational Neuroscience*, 30(1), 45–67.
- Wieser, H. G. (2004). ILAE Commission Report. Mesial temporal lobe epilepsy with hippocampal sclerosis. *Epilepsia*, 45(6), 695–714.

A Acknowledgement of generative-AI use

In line with the University of Sydney’s policy on generative AI, the author used Anthropic Claude (Opus 4.x) for language editing, structural suggestions, and assembling pre-computed result figures into report form, and AI-assisted coding tools (Cursor, Claude Code) for code-completion support while developing some of the MATLAB analysis scripts and their documentation. These tools were *not* used to choose the research question, define the methodology, select parameters, decide which results to report, perform the statistical interpretation, or generate unsupported scientific claims. All scientific design, parameter choices, result validation, and interpretation were performed by the author, who takes full responsibility for the submitted work and confirms it is their own, prepared in accordance with University guidelines.

B Reproducibility

All results derive from three configuration-driven pipeline runs (`column`, `parkes`, `tvb`) over the six-mouse cohort, executed in MATLAB R2024b. Each run records its full parameter set and a step-by-step manifest, and writes per-mouse stability results, cohort summaries, strain contrasts, laterality tests, and network- D_{st} tables. Numerical settings: continuous-time OU dynamics, covariance power-series cap 10^8 ; Epileptor $\tau_0 = 6667$, healthy $x_0 = -2.3$, $x_{0,i}^c$ bisection tolerance 10^{-4} ; Bonferroni $\alpha = 0.05$.

C Full result figures

This appendix reproduces *every* figure generated by the three configuration-driven pipeline runs of the latest experiment configuration (`column`, `parkes`, `tvb`; run 2026-05-27) over the six-mouse cohort. A representative subset appears in the main text; the complete set is collected here for reference. Within each panel grid the two ANDERSON case mice precede the four ARNOLD controls.

C.1 Network-level D_{st} across the cohort

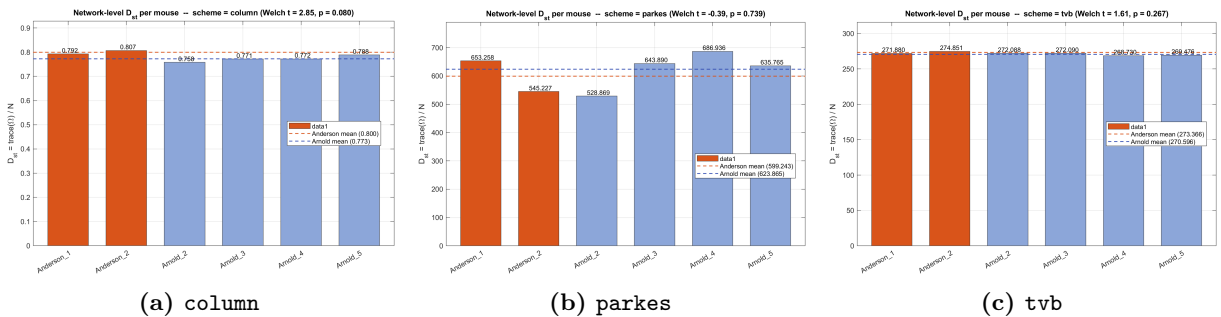


Figure 7: Per-mouse network $D_{st} = \text{trace}(\Omega)/N$ for each normalisation scheme. Only `column` separates the strains; under `parkes`/`tvb` the cohort ranges overlap.

C.2 Per-region stability centralities

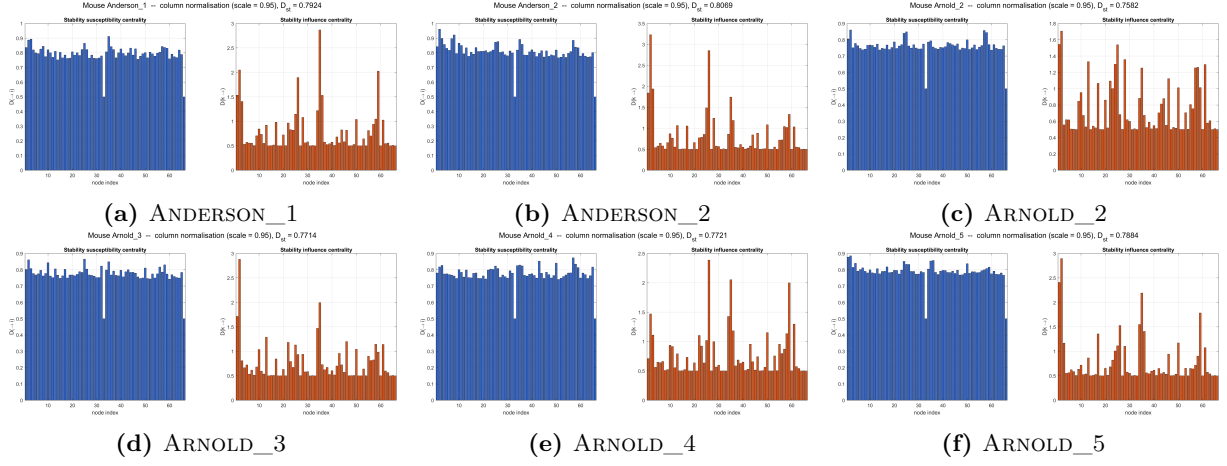


Figure 8: Per-region susceptibility $D(\rightarrow i)$ and influence $D(k \rightarrow)$ for all six mice, column scheme.

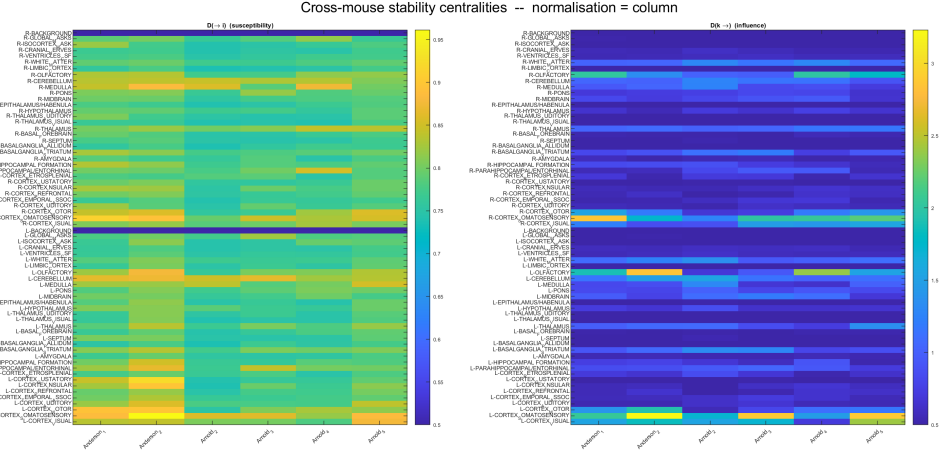


Figure 9: Cohort overview of stability centralities, column scheme.

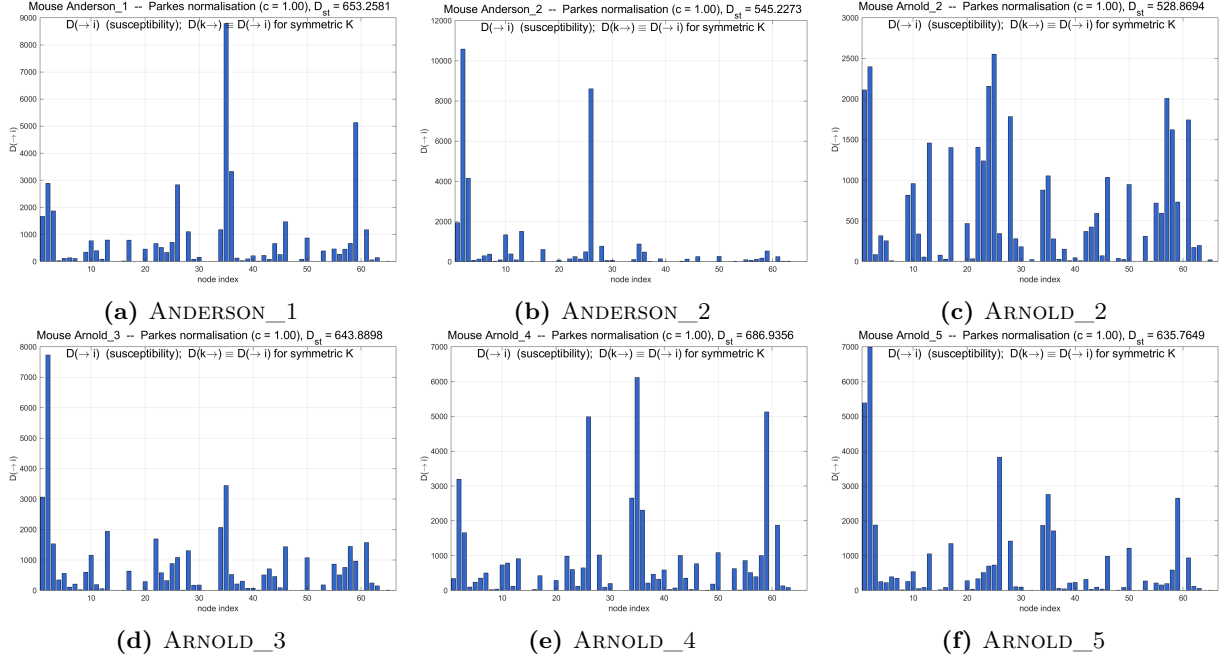


Figure 10: Per-region susceptibility $D(\rightarrow i)$ and influence $D(k \rightarrow)$ for all six mice, parkes scheme.

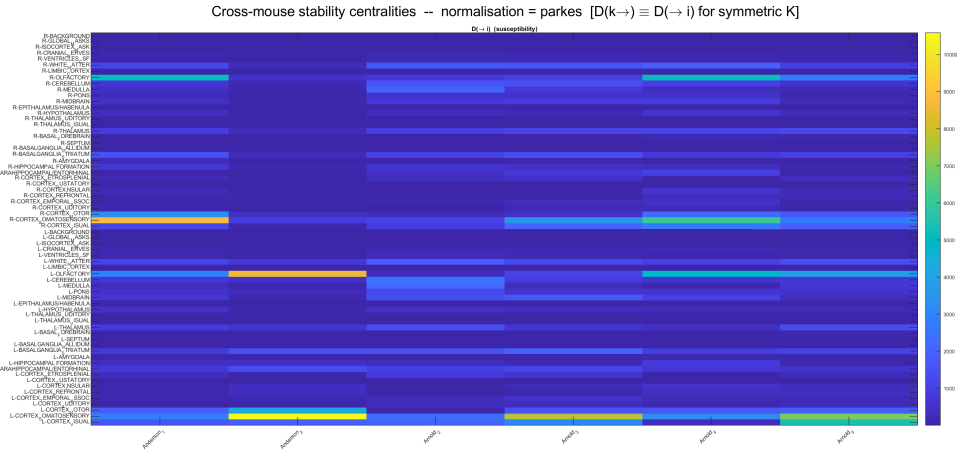


Figure 11: Cohort overview of stability centralities, parkes scheme.

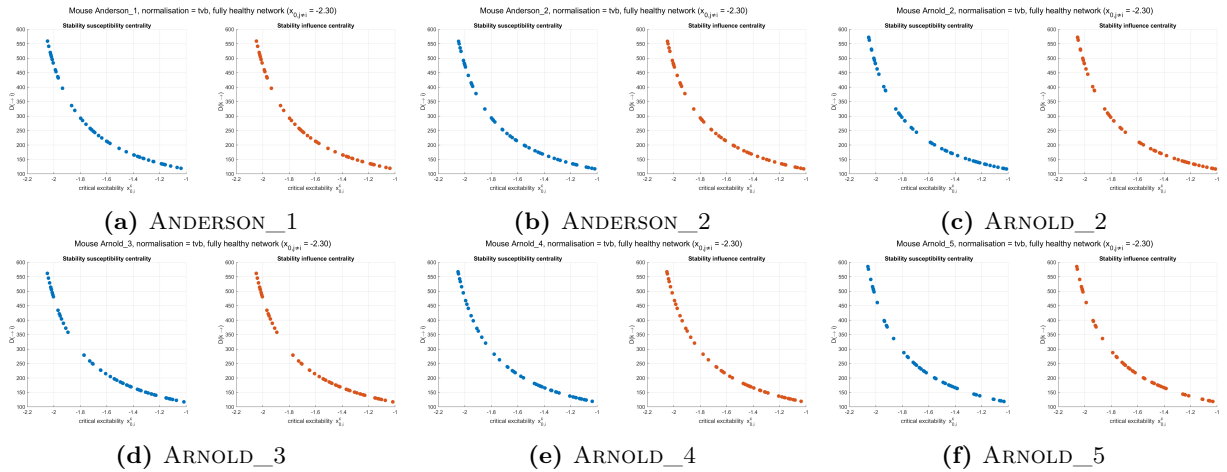


Figure 12: Per-region susceptibility $D(\rightarrow i)$ and influence $D(k \rightarrow)$ for all six mice, tvb scheme.

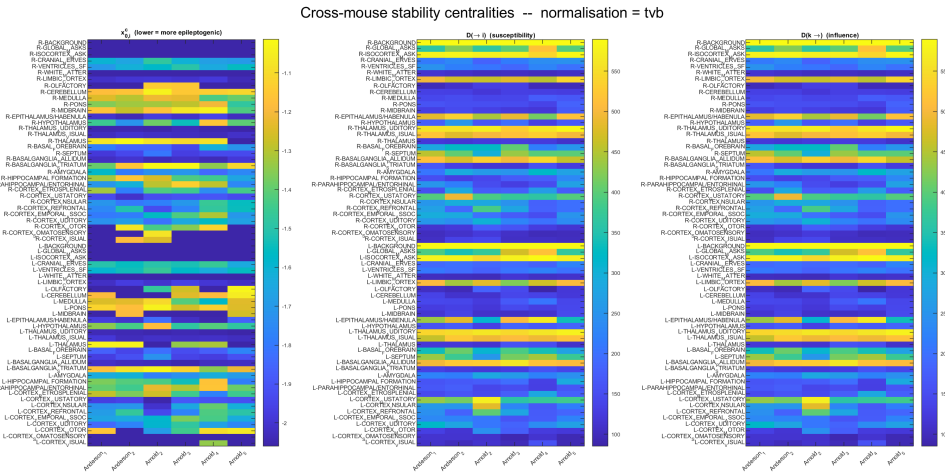


Figure 13: Cohort overview of stability centralities, tvb scheme.

C.3 Centrality correlations

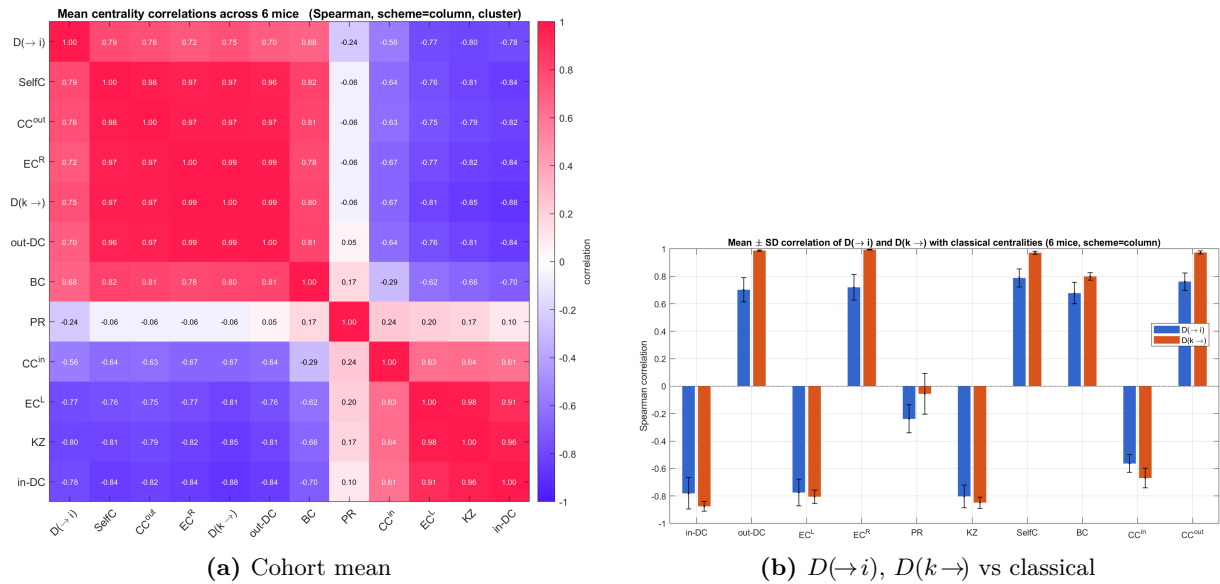


Figure 14: Centrality correlation summary, column scheme.

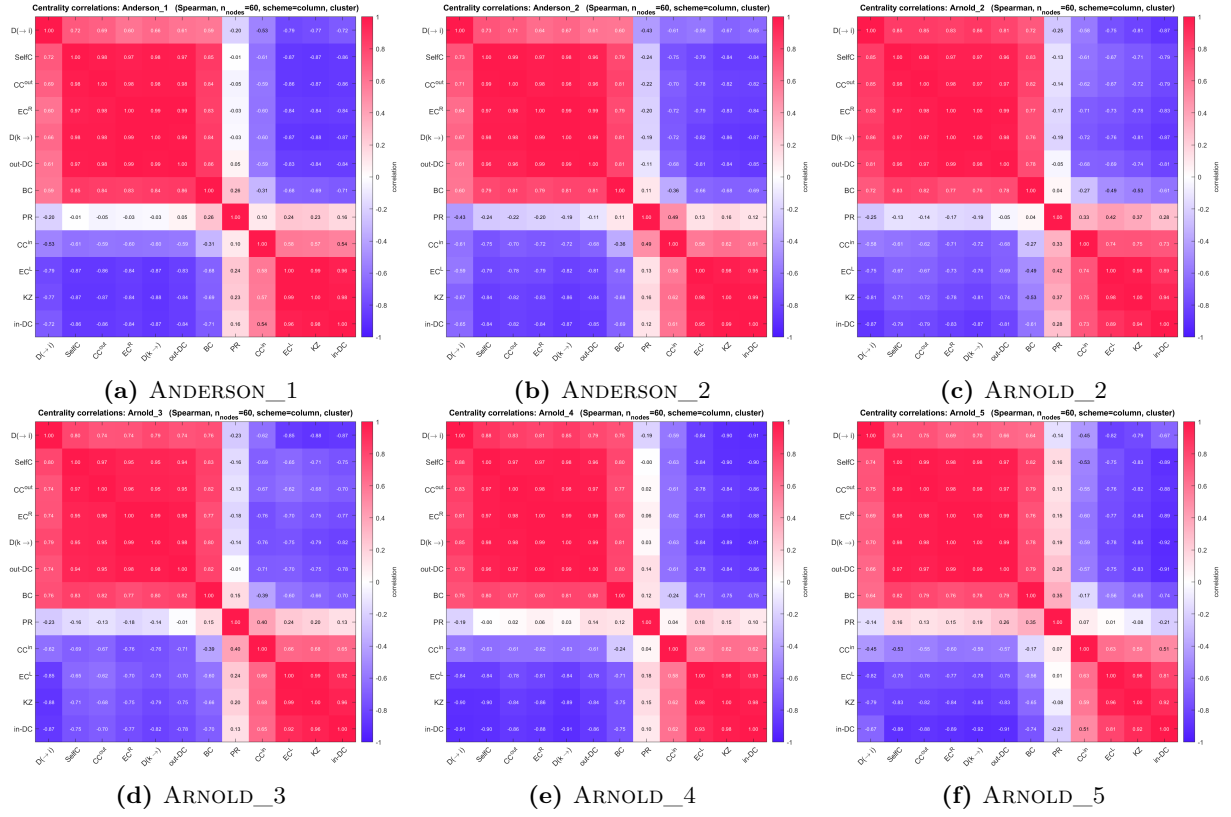


Figure 15: Per-mouse centrality correlation matrices, column scheme.

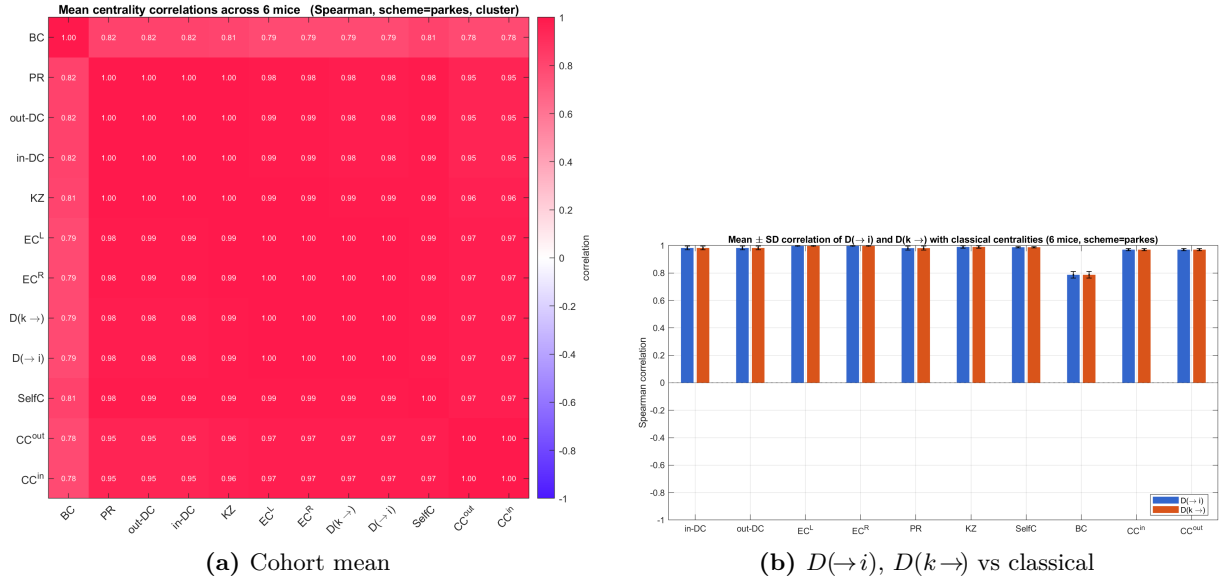


Figure 16: Centrality correlation summary, parkes scheme.

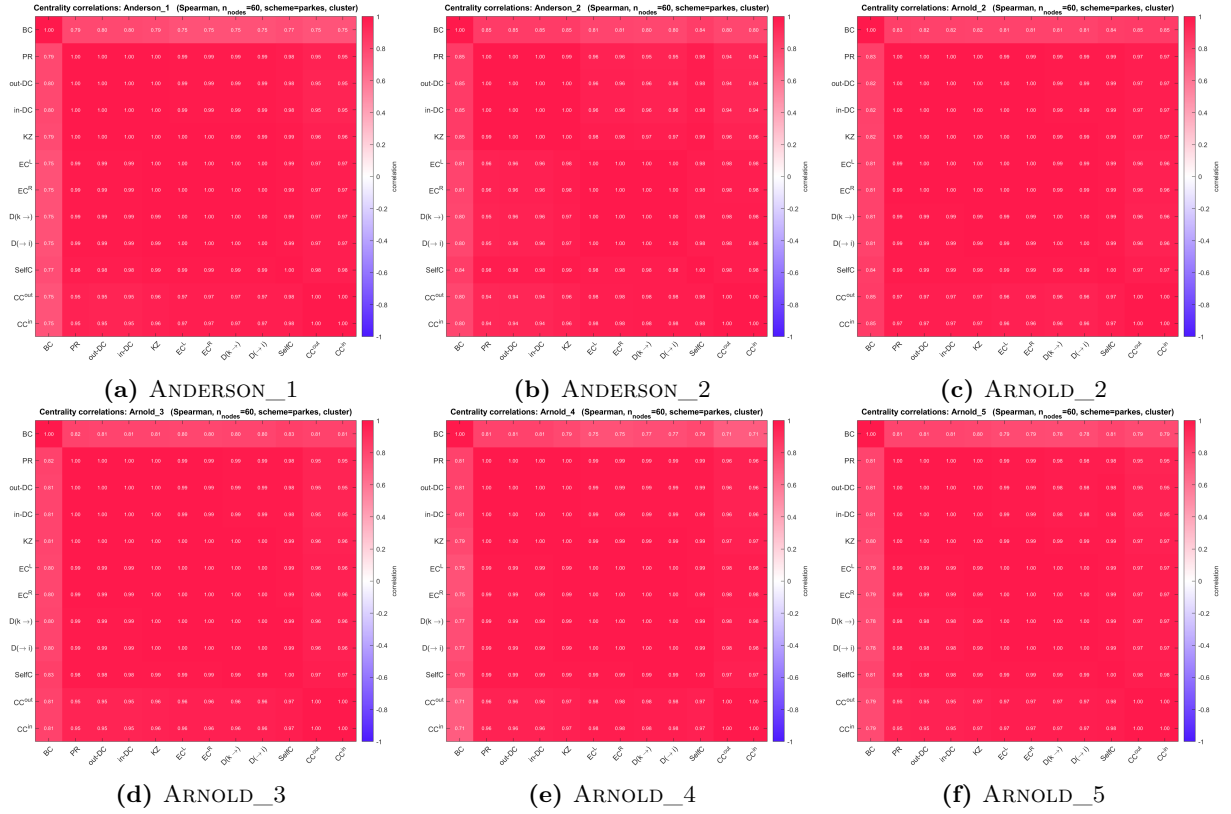


Figure 17: Per-mouse centrality correlation matrices, parkes scheme.

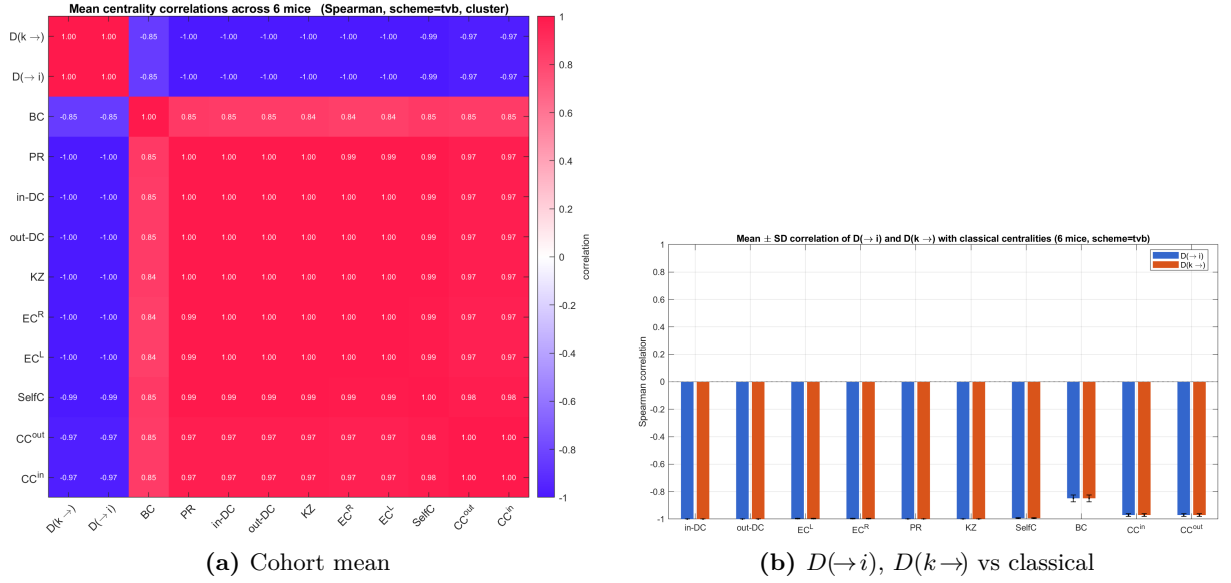


Figure 18: Centrality correlation summary, tvb scheme.

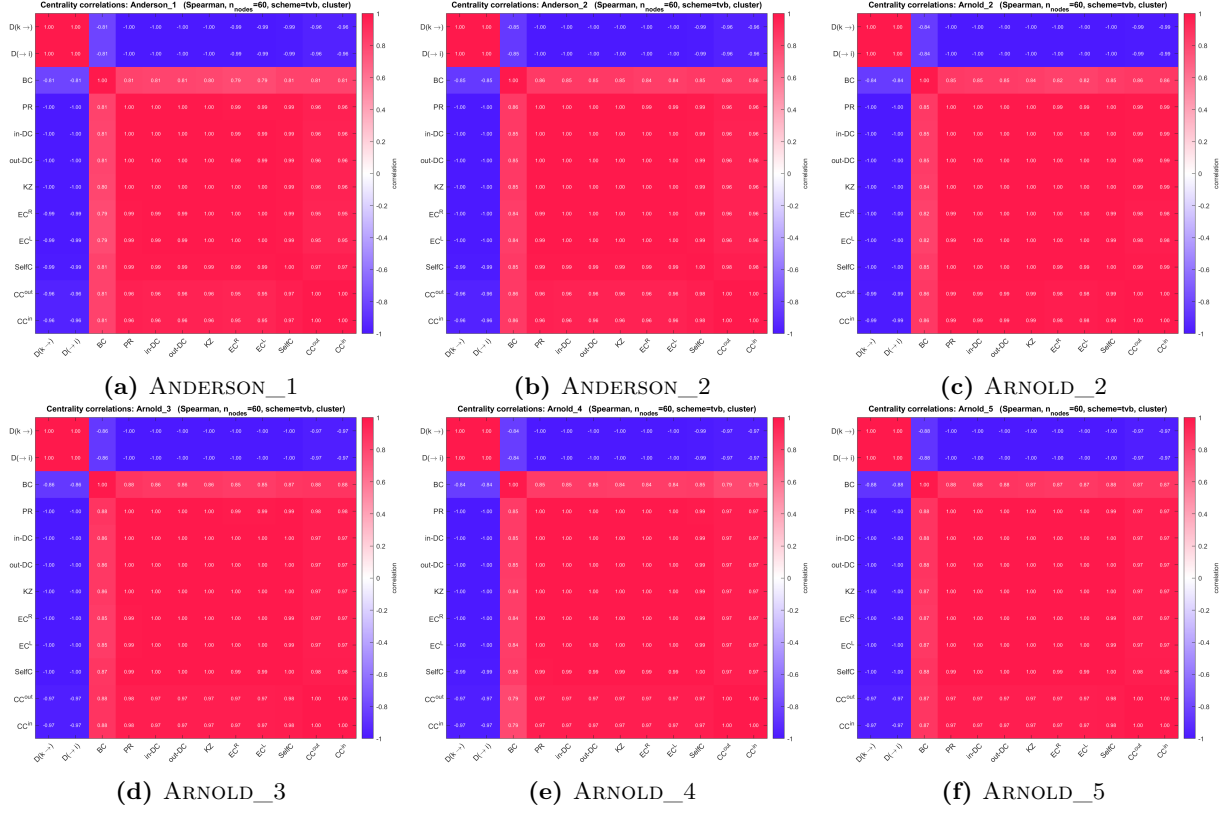


Figure 19: Per-mouse centrality correlation matrices, **tvb** scheme.

C.4 Anderson vs Arnold per-region contrasts

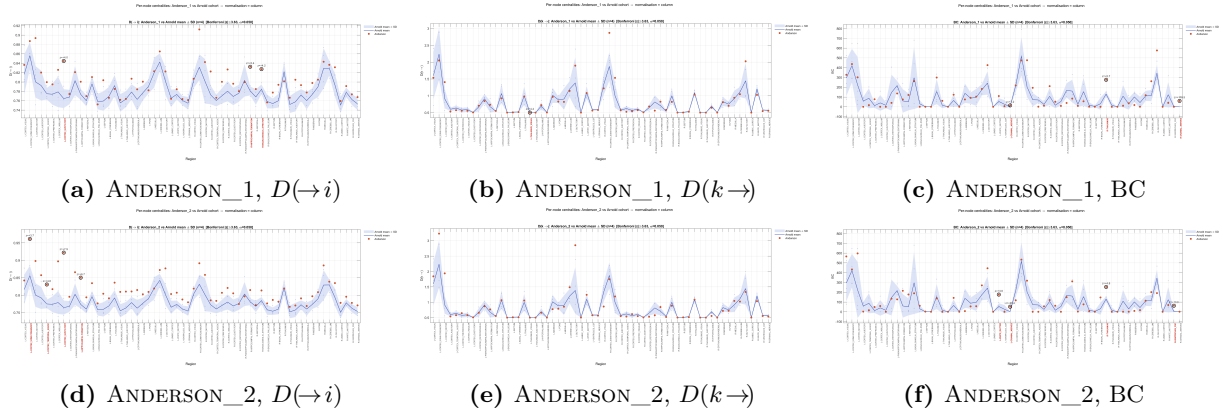


Figure 20: Each case mouse against the four-control band, column scheme: susceptibility $D(\rightarrow i)$, influence $D(k \rightarrow)$, and betweenness BC.

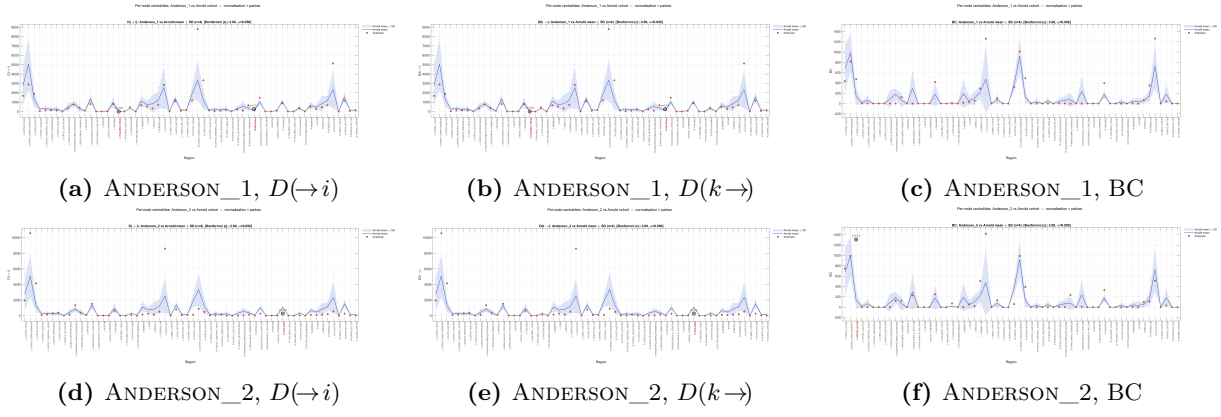


Figure 21: Each case mouse against the four-control band, **parkes** scheme: susceptibility $D(\rightarrow i)$, influence $D(k \rightarrow)$, and betweenness BC.

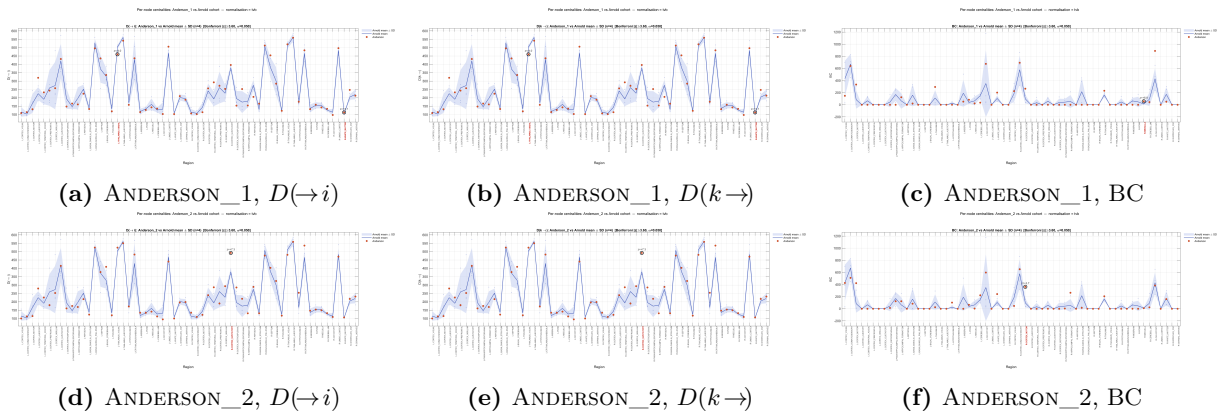


Figure 22: Each case mouse against the four-control band, **tvb** scheme: susceptibility $D(\rightarrow i)$, influence $D(k \rightarrow)$, and betweenness BC.

C.5 Left-right asymmetry

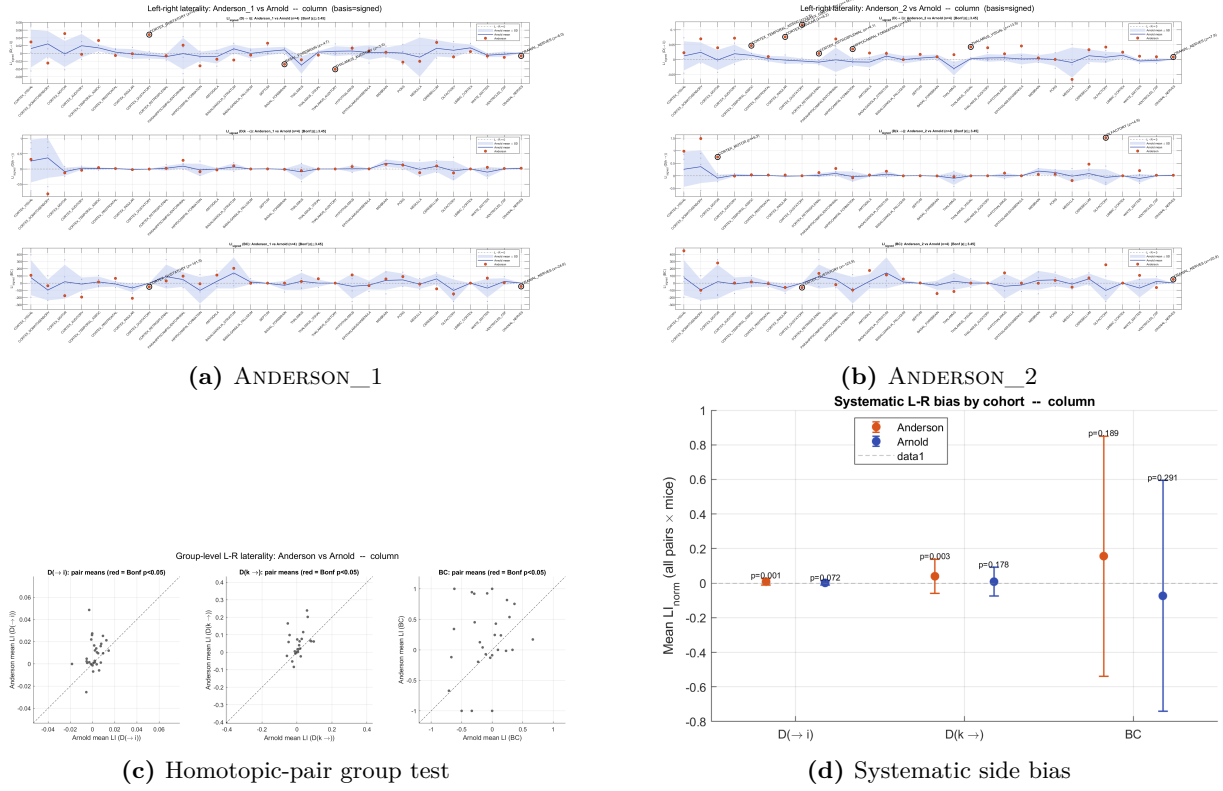


Figure 23: Left-right asymmetry views, column scheme.

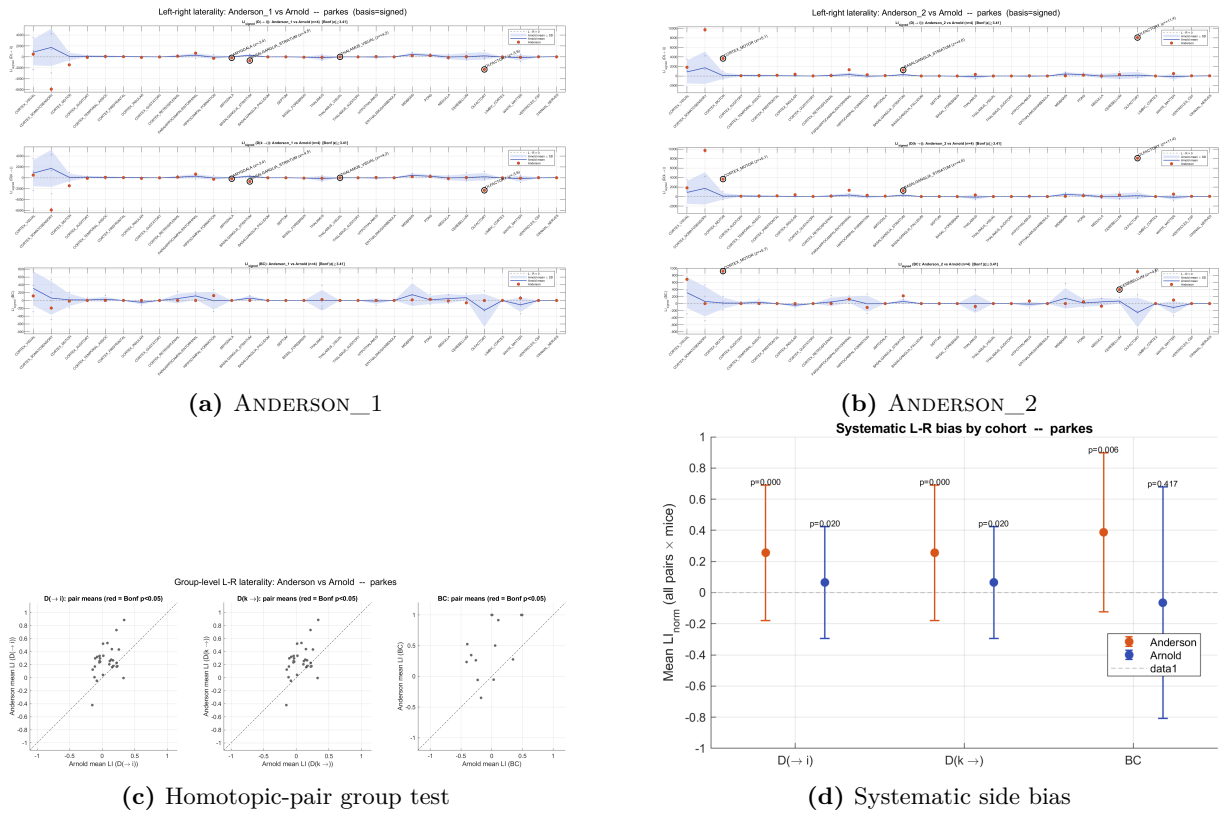


Figure 24: Left-right asymmetry views, parkes scheme.

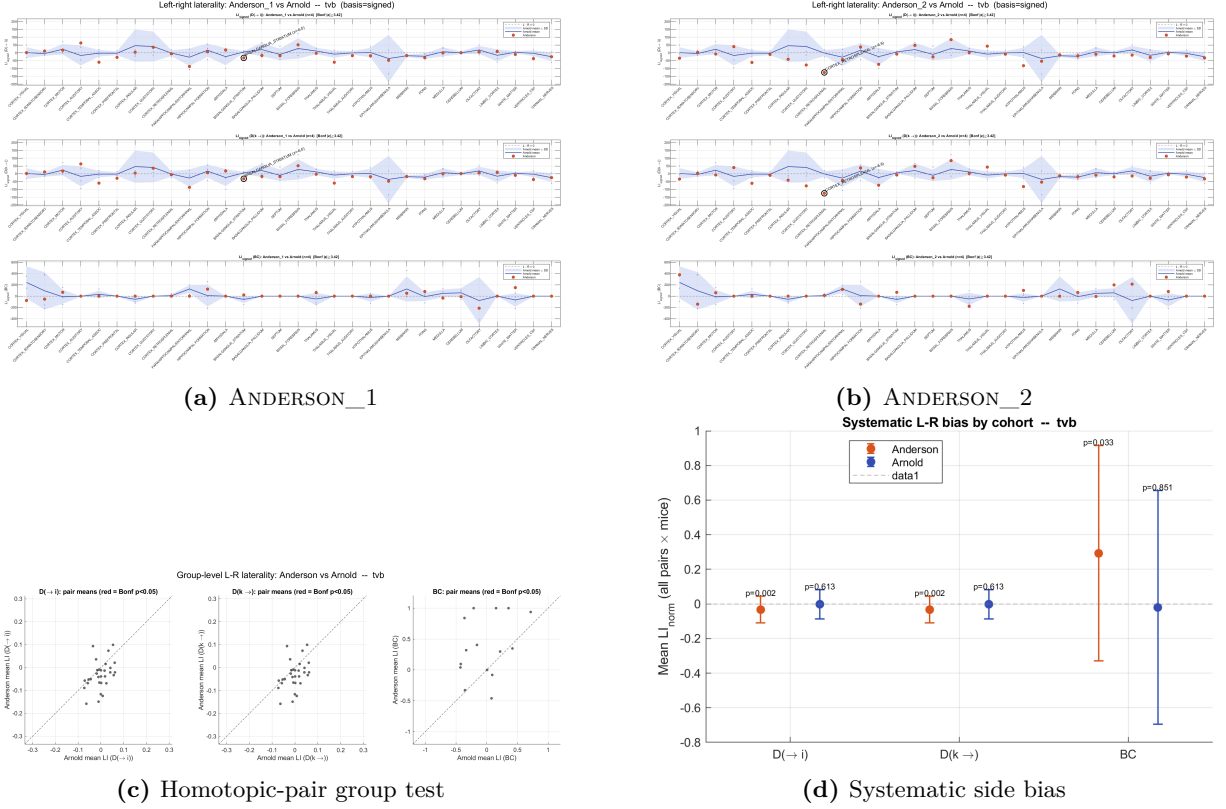


Figure 25: Left-right asymmetry views, tvb scheme.

C.6 Cohort comparison figures (column)

These two summary figures also appear in the main text (Fig. 5, Fig. 6).

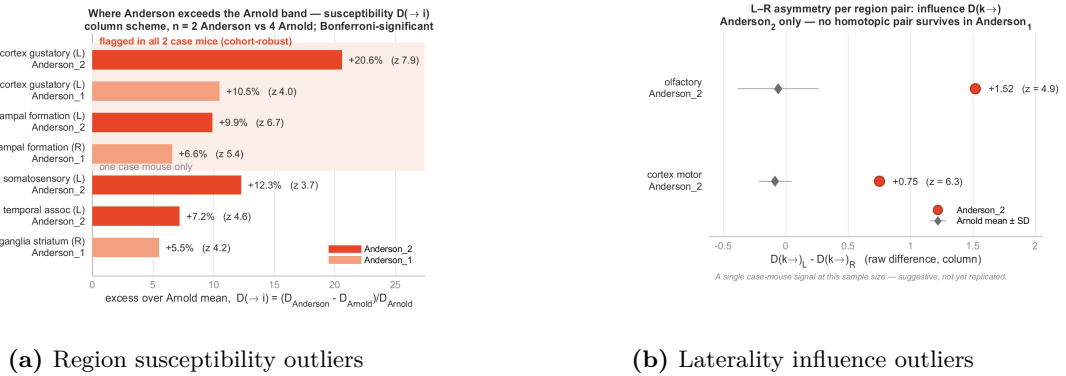


Figure 26: Cohort-level Bonferroni-significant outliers, column scheme.

D Full result tables

Numeric summaries underlying the figures above. Complete per-node CSVs for every run are written to the run folders by the pipeline (see Appendix B).

D.1 Network D_{st} and spectral radius $\rho(C)$

Table 3: Per-mouse D_{st} and $\rho(C)$ for the three schemes. Case mice (ANDERSON) above the rule.

Mouse	D_{st}			$\rho(C)$		
	column	parkes	tvb	column	parkes	tvb
Anderson_1	0.7924	653.3	271.9	0.95	0.999988	0.999152
Anderson_2	0.8069	545.2	274.9	0.95	0.999986	0.999152
Arnold_2	0.7582	528.9	272.1	0.95	0.999986	0.999152
Arnold_3	0.7714	643.9	272.1	0.95	0.999988	0.999152
Arnold_4	0.7721	686.9	268.7	0.95	0.999989	0.999152
Arnold_5	0.7884	635.8	269.5	0.95	0.999988	0.999152

D.2 Top regions by stability susceptibility

Table 4: Top 15 non-trivial regions by mean susceptibility rank, column scheme (lower rank = more central).

#	Region	mean rank $D(\rightarrow i)$	mean rank $D(k \rightarrow)$
1	L-CORTEX_SOMATOSENSORY	3	1.67
2	R-CORTEX_SOMATOSENSORY	4.33	3.83
3	L-CEREBELLUM	5.5	8.5
4	R-CEREBELLUM	8.17	11.2
5	R-MEDULLA	9	16
6	L-CORTEX_VISUAL	11.3	6.83
7	L-CORTEX_MOTOR	12	14.7
8	R-THALAMUS	12	10.8
9	L-MEDULLA	14.3	20.3
10	L-OLFACTORY	14.5	9.33
11	L-PARAHIPPOCAMPAL/ENTORHINAL	15	16.5
12	R-CORTEX_VISUAL	15.5	9.67
13	L-BASALGANGLIA_STRIATUM	16.2	12.7
14	R-OLFACTORY	16.5	7.33
15	R-CORTEX_MOTOR	16.7	13.3

Table 5: Top 15 non-trivial regions by mean susceptibility rank, **parkes** scheme (lower rank = more central).

#	Region	mean rank $D(\rightarrow i)$	mean rank $D(k \rightarrow)$
1	L-CORTEX_SOMATOSENSORY	2.17	2.17
2	R-CORTEX_SOMATOSENSORY	4.67	4.67
3	L-CORTEX_VISUAL	8.33	8.33
4	L-OLFACTORY	8.67	8.67
5	R-OLFACTORY	8.83	8.83
6	L-WHITE_MATTER	9.17	9.17
7	L-BASALGANGLIA_STRIATUM	9.67	9.67
8	R-WHITE_MATTER	10.5	10.5
9	L-CORTEX_MOTOR	11.3	11.3
10	R-CORTEX_VISUAL	11.5	11.5
11	L-CEREBELLUM	13	13
12	R-THALAMUS	13	13
13	R-BASALGANGLIA_STRIATUM	13.2	13.2
14	R-CEREBELLUM	13.8	13.8
15	L-PARAHIPPOCAMPAL/ENTORHINAL	14	14

Table 6: Top 15 non-trivial regions by mean susceptibility rank, **tvb** scheme (lower rank = more central).

#	Region	mean rank $D(\rightarrow i)$	mean rank $D(k \rightarrow)$	mean rank $x_{0,i}^c$
1	R-THALAMUS_AUDITORY	5.17	5.17	1.83
2	L-THALAMUS_AUDITORY	5.67	5.67	2.17
3	L-BASALGANGLIA_PALLIDUM	9	9	4.83
4	R-THALAMUS_VISUAL	9.67	9.67	5.67
5	R-BASALGANGLIA_PALLIDUM	10	10	6.17
6	L-THALAMUS_VISUAL	10.7	10.7	6.33
7	R-EPITHALAMUS/HABENULA	11.8	11.8	7.5
8	R-LIMBIC_CORTEX	12.2	12.2	7.83
9	L-LIMBIC_CORTEX	12.8	12.8	8.33
10	L-EPITHALAMUS/HABENULA	13.7	13.7	9.17
11	L-CORTEX_GUSTATORY	15.8	15.8	11.2
12	R-CORTEX_GUSTATORY	17	17	12.2
13	R-SEPTUM	17.5	17.5	12.5
14	L-SEPTUM	19.3	19.3	14.2
15	L-BASAL_FOREBRAIN	20.8	20.8	15

D.3 Anderson vs Arnold per-region outliers

Table 7: Flagged per-region contrasts ($|z| \geq 2$) vs the control band, **column** scheme. p_b = Bonferroni-corrected p ; * marks significance at $\alpha = 0.05$.

Mouse	Region	z_{susc}	p_b^{susc}	z_{infl}	p_b^{infl}	z_{BC}	p_b^{BC}
Anderson_1	R-CRANIAL_NERVES	1.12	1	-0.24	1	119.50	$< 10^{-15}$ *
Anderson_1	L-CRANIAL_NERVES	0.51	1	-0.17	1	7.17	1.3×10^{-10} *
Anderson_1	R-THALAMUS	-1.04	1	-0.70	1	5.65	2.8×10^{-6} *
Anderson_1	R-HIPPOCAMPAL FORMATION	5.38	1.3×10^{-5} *	0.90	1	-0.66	1
Anderson_1	L-THALAMUS_VISUAL	0.35	1	4.42	0.00172*	1.50	1
Anderson_1	R-BASALGANGLIA_STRIATUM	4.25	0.00379*	-0.90	1	-0.69	1
Anderson_1	L-CORTEX_GUSTATORY	4.00	0.0111*	-0.32	1	-0.50	1
Anderson_2	L-CRANIAL_NERVES	1.88	1	-0.86	1	29.42	$< 10^{-15}$ *
Anderson_2	R-VENTRICLES_CSF	1.33	1	-2.32	1	10.77	$< 10^{-15}$ *
Anderson_2	L-CORTEX_GUSTATORY	7.87	6.3×10^{-13} *	0.12	1	-0.50	1
Anderson_2	L-HIPPOCAMPAL FORMATION	6.75	2.7×10^{-9} *	0.53	1	1.29	1
Anderson_2	R-THALAMUS	-0.15	1	-0.12	1	4.80	2.9×10^{-4} *
Anderson_2	L-CORTEX_TEMPORAL_ASSOC	4.57	8.8×10^{-4} *	-0.91	1	-0.70	1
Anderson_2	L-CORTEX_SOMATOSENSORY	3.74	0.033*	1.31	1	0.07	1
Anderson_2	L-WHITE_MATTER	3.03	0.433	0.77	1	3.65	0.0469*

Table 8: Flagged per-region contrasts ($|z| \geq 2$) vs the control band, **parkes** scheme. p_b = Bonferroni-corrected p ; * marks significance at $\alpha = 0.05$.

Mouse	Region	z_{susc}	p_b^{susc}	z_{infl}	p_b^{infl}	z_{BC}	p_b^{BC}
Anderson_1	R-AMYGDALA	7.40	2.2×10^{-11} *	7.40	2.2×10^{-11} *	—	1
Anderson_1	L-THALAMUS_VISUAL	3.86	0.0174*	3.86	0.0174*	—	1
Anderson_2	R-THALAMUS	-7.45	1.5×10^{-11} *	-7.45	1.5×10^{-11} *	2.10	1
Anderson_2	L-CORTEX_MOTOR	3.51	0.0692	3.51	0.0692	7.43	1.7×10^{-11} *

Table 9: Flagged per-region contrasts ($|z| \geq 2$) vs the control band, **tvb** scheme. p_b = Bonferroni-corrected p ; * marks significance at $\alpha = 0.05$.

Mouse	Region	z_{susc}	p_b^{susc}	z_{infl}	p_b^{infl}	z_{BC}	p_b^{BC}
Anderson_1	R-MEDULLA	-0.67	1	-0.67	1	6.78	1.8×10^{-9} *
Anderson_1	R-WHITE_MATTER	4.15	0.00527*	4.15	0.00527*	-1.01	1
Anderson_1	L-THALAMUS_VISUAL	-3.98	0.011*	-3.98	0.011*	—	1
Anderson_2	R-CORTEX_GUSTATORY	7.29	4.9×10^{-11} *	7.29	4.9×10^{-11} *	—	1
Anderson_2	R-CORTEX_MOTOR	-0.79	1	-0.79	1	3.70	0.0345*

D.4 Systematic laterality

Table 10: Cohort-wide sign-rank test of the normalised laterality index LI_{norm} against zero, all schemes. Side = inferred dominant hemisphere (nominal labels).

Scheme	Cohort	Metric	mean LI	sd LI	sign-rank p	Side
column	Anderson	D_susc	0.00879	0.0188	7.9×10^{-4}	L
column	Arnold	D_susc	0.00173	0.0131	0.0724	none
column	Anderson	D_infl	0.0399	0.0982	0.0026	L
column	Arnold	D_infl	0.00862	0.083	0.178	none
column	Anderson	BC	0.156	0.694	0.189	none
column	Arnold	BC	-0.0734	0.668	0.291	none
parkes	Anderson	D_susc	0.256	0.435	1.0×10^{-4}	L
parkes	Arnold	D_susc	0.0654	0.359	0.0203	L
parkes	Anderson	D_infl	0.256	0.435	1.0×10^{-4}	L
parkes	Arnold	D_infl	0.0654	0.359	0.0203	L
parkes	Anderson	BC	0.387	0.511	0.00609	L
parkes	Arnold	BC	-0.0651	0.744	0.417	none
tvb	Anderson	D_susc	-0.0332	0.0769	0.00151	R
tvb	Arnold	D_susc	-0.00186	0.0838	0.613	none
tvb	Anderson	D_infl	-0.0332	0.0769	0.00151	R
tvb	Arnold	D_infl	-0.00186	0.0838	0.613	none
tvb	Anderson	BC	0.292	0.622	0.033	L
tvb	Arnold	BC	-0.0203	0.676	0.851	none

D.5 Homotopic-pair laterality tests

Table 11: Per-pair Welch t -test between strains, column scheme: 10 smallest Bonferroni p . ✓ marks $p_b < 0.05$.

Region	Metric	LI_{And}	LI_{Arn}	t	p_b	sig.
CORTEX_VISUAL	D_susc	0.016	0.00737	0.58	1	
CORTEX_SOMATOSENSORY	D_susc	0.0119	0.0146	-0.10	1	
CORTEX_MOTOR	D_susc	0.0261	-0.000446	2.45	1	
CORTEX_AUDITORY	D_susc	0.0213	0.0125	0.36	1	
CORTEX_TEMPORAL_ASSOC	D_susc	0.0252	0.00942	3.27	1	
CORTEX_PREFRONTAL	D_susc	0.00152	0.00215	-0.11	1	
CORTEX_INSULAR	D_susc	0.022	-0.00112	1.01	1	
CORTEX_GUSTATORY	D_susc	0.0486	-0.00299	2.68	1	
CORTEX_RETROSPLINIAL	D_susc	0.00464	-0.00525	1.12	1	
PARAHIPPOCAMPAL/ENTORHINAL	D_susc	0.0274	-0.000259	1.46	1	

Table 12: Per-pair Welch t -test between strains, **parkes** scheme: 10 smallest Bonferroni p . ✓ marks $p_b < 0.05$.

Region	Metric	LI _{And}	LI _{Arn}	t	p_b	sig.
CORTEX_VISUAL	D_susc	0.533	0.0803	0.98	1	
CORTEX_SOMATOSENSORY	D_susc	0.17	0.223	-0.08	1	
CORTEX_MOTOR	D_susc	0.256	-0.0398	0.51	1	
CORTEX_AUDITORY	D_susc	-0.00396	0.328	-0.50	1	
CORTEX_TEMPORAL_ASSOC	D_susc	0.732	0.219	1.99	1	
CORTEX_PREFRONTAL	D_susc	0.256	0.117	1.18	1	
CORTEX_INSULAR	D_susc	0.321	-0.0774	0.57	1	
CORTEX_GUSTATORY	D_susc	0.298	-0.107	0.58	1	
CORTEX_RETROSPLLENIAL	D_susc	0.523	0.0136	1.48	1	
PARAHIPPOCAMPAL/ENTORHINAL	D_susc	0.887	0.337	2.41	1	

Table 13: Per-pair Welch t -test between strains, **tvb** scheme: 10 smallest Bonferroni p . ✓ marks $p_b < 0.05$.

Region	Metric	LI _{And}	LI _{Arn}	t	p_b	sig.
CORTEX_VISUAL	D_susc	-0.0672	0.00111	-0.67	1	
CORTEX_SOMATOSENSORY	D_susc	0.0358	-0.0212	1.45	1	
CORTEX_MOTOR	D_susc	0.021	0.0605	-0.61	1	
CORTEX_AUDITORY	D_susc	0.0934	-0.0349	1.88	1	
CORTEX_TEMPORAL_ASSOC	D_susc	-0.116	-0.000611	-4.30	1	
CORTEX_PREFRONTAL	D_susc	-0.0413	-0.011	-1.61	1	
CORTEX_INSULAR	D_susc	-0.0333	0.0562	-1.09	1	
CORTEX_GUSTATORY	D_susc	-0.0211	0.0412	-0.76	1	
CORTEX_RETROSPLLENIAL	D_susc	-0.149	-0.0104	-1.04	1	
PARAHIPPOCAMPAL/ENTORHINAL	D_susc	-0.158	-0.0639	-1.12	1	